NE 763 Exam 2

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1) SFRs offer a variety of safety benefits. Sodium, as a coolant, has a very high thermal conductivity and boiling point. At atmospheric pressure, sodium has a wide temperature range where it exists as a liquid which allows the reactor to operate at atmospheric pressure. The high boiling point and thermal conductivity allows for the coolant to accept a ton of heat without boiling, allowing the reactor to shut itself down using thermal inertia, even without operator intervention with control rods in some cases.

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2) Uranium fuels have multiple phases.

The α-phase has an orthorhombic structure known for highly anisotropic swelling under irradiation. Adding Zr to uranium helps to stabilize the γ-uranium phase which has more isotropic swelling properties. Additionally, alloying Zr helps to raise the melting point of the fuel which increases safety margins. Finally, alloying with Zr forms a protective rind on the outside of a fuel slug which prevents FCCI, or elemental diffusion between fuel and cladding.

3) Metallic fuel systems swell quite a bit. In the beginning of life, fuel swells roughly 30% in the first 2% of burnup. As fission product gases form, the fuel swells which can cause stress on the cladding and FCCI which leads to degraded material performance. By lowering the smear density, the fuel is allowed to swell without interacting with the cladding. Fission product gases form voids that connect and reach the edge of the fuel. The gas plenum serves as a space for the fission product gases to collect

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upon exiting the fuel. Bond sodium then fills the void space which recovers thermal conductivity.

4) In metallic fuels, constituents redistribute into 3 zones. A Zr-rich central zone, a U-rich intermediate zone, and a Zr-rich zone or the outside of the fuel pin. At high temperatures, such as in the middle of the fuel pin (radially), the γ-uranium phase dominates. Zr is more soluble in the γ phase which leads to a Zr-rich zone in the middle of the fuel pin. Since Zr collects in the middle of the pin, there is a uranium-rich zone outside of that. High Pu concentrations are known to accelerate constituent redistribution.

The impacts of constituent redistribution are that thermal conductivity and Phase behavior are different across the radial profile of the fuel slug. Swelling then becomes more anisotropic and faster in the radial direction than the axial direction.

- Soret diffusion also

5) In the initial phase of operation where fuel swells freely, thermal conductivity degrades with burnup. 8/8

This is because fission product gas formation leads to porosity in the fuel which has very poor thermal conductivity. When the fuel reaches maximum swelling, fission gas porosity networks connect and exit the fuel releasing gas to the gas plenum on top of the fuel slug. Bond Sodium fills the porosity gaps which then improves thermal conductivity.

b) FCCI, or Fuel cladding Chemical Interaction is a diffusion process where constituents of the fuel and cladding react. It is often life-limiting for SFRs. Primarily, lanthanides such as cerium and ~~niobium~~ Nd travel from the fuel into the cladding. These elements react with cladding to form precipitates in the cladding leading to material attack and wastage. Iron (Fe + Cr) is the primary species diffusing from the cladding into the fuel. Iron forms alloys with fuel that have lower melting points leading to limited operational temperatures and decreased efficiency.

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It should be noted that Zr is beneficial for FCCI because it forms a rind on the outside of the fuel pin which serves as a barrier to diffusion.

7) MOX fuels have steep radial temperature gradients. Quickly after the beginning of operations, an evaporation - condensation mechanism begins where fuel evaporates on the hot side of a pore and condenses on the cold side, effectively allowing pores to climb up the temperature gradient. As pores climb, they collect and form a void in the middle of the fuel pin. The paths forged by the pores to the center of the fuel align the grains to form a columnar grain region. Outside of the columnar grain region, temperatures are still hot enough to cause grain growth, but not hot enough to cause grain

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aligning motion. This zone is called the equiaxed grain zone. Finally, outside the equiaxed grain zone, there is an as-fabricated zone where temps are too low to significantly change the micro structure.

8) In a MOX fuel pin, Uranium oxide is more volatile than plutonium oxide.

9/10 Uranium oxides evaporate and condense sequentially causing plutonium to be left behind in a ring surrounding the void in a MOX pin. The evaporation - condensation process causes oxygen to migrate down the thermal gradient leaving very little oxygen near the core of the fuel pin, and roughly Stoichiometric oxygen in the outside region of the fuel pin.

oxygen can move via Soret effects

- Pu agglomerates

9) JOG, or the Joint Oxide Gaine, is a layer of dense oxides that build up in the region between the fuel and the cladding. It's primarily composed of fission product oxides. JOG occurs because volatile fission products evaporate in the hot part of fuel and the condense in the cold part of the fuel forming oxides such as cesium molybdates that make up the JOG layer. JOG is good for thermal conductivity because it has higher conductivity than gases in the space between the fuel and the cladding.

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Additionally, JOG serves as a buffer between the fuel and cladding to prevent mechanical interaction and JOG helps reduce swelling since volatile fission products solidify in the JOG layer. Unfortunately, JOG is bad for FCCI as fission products directly contact the cladding and cause corrosion.

10) Sodium corrosion has 2 modes. In mode 1, elements are dissolved in the sodium from cladding, specifically Ni and Mn at the beginning of operation, and Fe at steady state operation. Solubility is temperature dependent, so typically elements dissolve into the sodium in the core where its hot and plate out on the much cooler heat exchanger. In mode 2, impurities in the sodium such as C, N, or O form compounds with sodium that react with cladding to cause corrosion and material wastage.

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- increase solubility